

Check up

- The voltage across a forward-biased silicon diode is approximately
(a) 0.7V (b) 0.3V (c) 0V (d) dependent on the bias voltage

Answer (a)

- In Figure 1, identify the forward-biased diode(s).

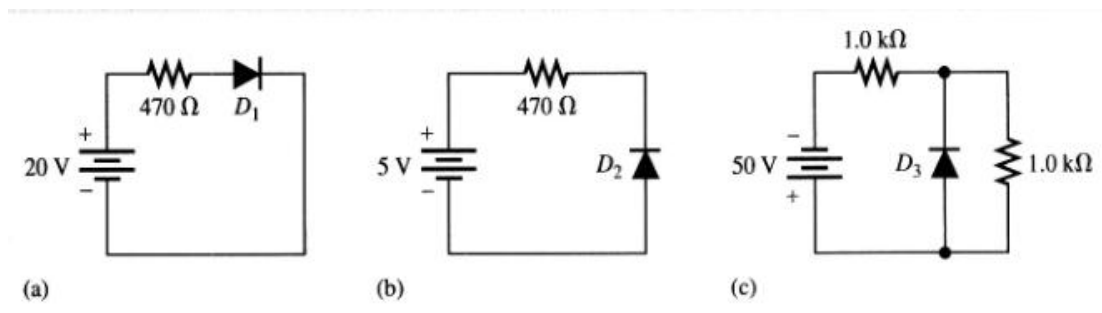


Figure 1

Answer (a) (c)

- When the positive lead of an analog ohmmeter is connected to the cathode of a diode and the negative lead is connected to the anode, the meter reads
(a) A very low resistance (b) an extremely high resistance or open
(c) a high resistance initially, decreasing to about 100Ω
(d) a gradually increasing resistance

Answer (b)

- The output frequency of a full-wave rectifier with a 60Hz sinusoidal input is
(a) 30Hz (b) 60Hz (c) 120Hz (d) 0Hz

Answer (c)

- if a single diode in a center-tapped full wave rectifier opens, the output is
(a) 0V (b) half-wave rectified (c) increased in amplitude (d) unaffected

Answer (b)

- During the positive half-cycle of the input voltage in a bridge rectifier,
(a) one diode is forward biased (b) all diodes are forward biased
(c) all diodes are reverse biased (d) two diodes are forward biased

Answer (d)

- The process of changing a half-wave or a full-wave rectified voltage to a constant DC voltage is called

(a) filtering (b) AC to DC conversion (c) damping (d) ripple suppression

Answer (b)

- Zener diodes are widely used as

(a) current limiters (b) power distributors (c) voltage regulators (d) variable resistors

Answer (c)

Problems

- Sketch the waveforms for the load current and voltage for Figure 1. Show the peak value.

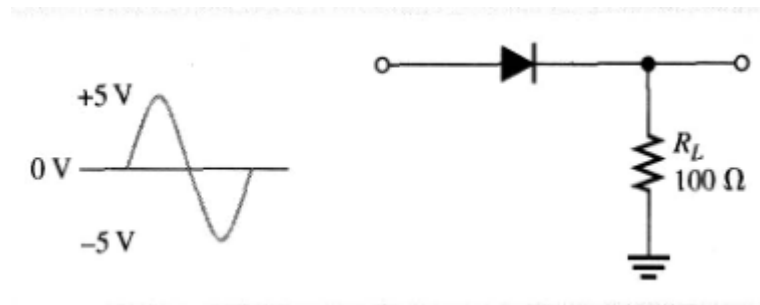


Figure 1

- Determine the peak voltage and peak power delivered to R_L in Figure 2.

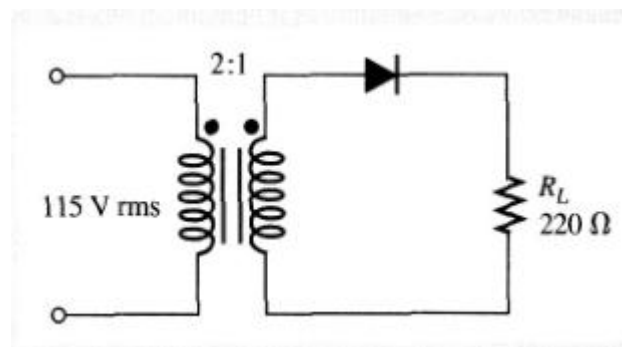


Figure 2

Answer 80.6V, 29.5W

- Figure 3 shows a zener diode regulator designed to hold 5.0V at the output. Assume the zener resistance is zero and the zener current ranges from 2mA minimum (I_{ZK}) to 30mA maximum (I_{ZM}). What are the minimum and maximum input source voltages for these current?

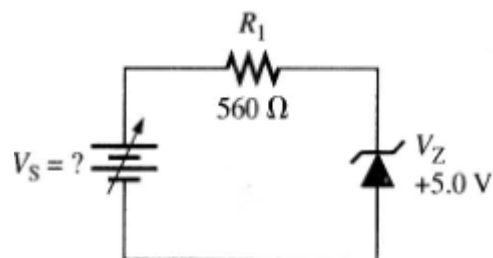


Figure 3

- From the meter readings in Figure 4, determine the most likely problem. State how you could quickly isolate the exact location of the problem.

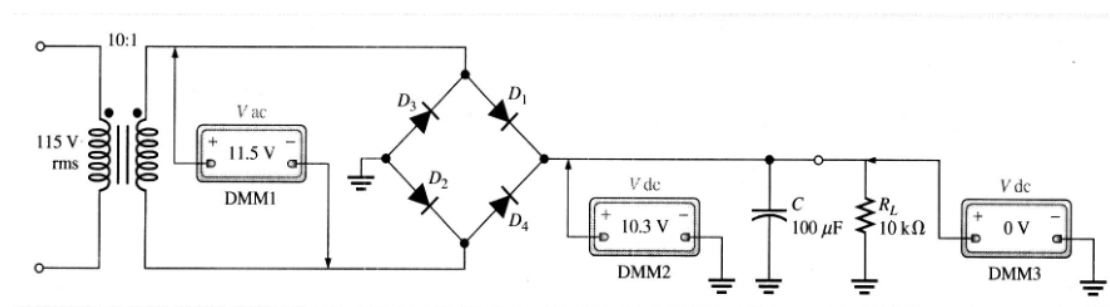


Figure 4

Answer 10k R_L resistor is disconnected.